

ActInfLab ~ 2021 Quarterly Roundtable #1

Roundtable | 2021 Quarterly Roundtable #1 | 1:21:56

hello

and welcome everyone to the active

inference lab

this is a special active inference live

stream it is march

2nd 2021 and this is our first quarterly

roundtable

for the active inference lap so this

should be a pretty fun

and unique stream welcome to the active

inference lab everyone

we're a participatory online lab that is

communicating

learning and practicing applied active

inference you can find us

on our website twitter through email our

youtube channel

our keybase or our discord teams which

we're going to go into

later in the stream so if you're curious

about what each of these different

platforms are for we're going to get

there

this is a recorded and an archived live

stream so please provide us with

feedback so that we can improve on our

work all backgrounds and perspectives

are welcome here and we will

use video etiquette netiquette for live

streams including muting

and using respectful speech behavior

let's just take

a step back and look at where we're at

on our calendar it's march 2nd and here

we are with our first quarterly

roundtable

it's been a pretty interesting year with

two months

of weekly discussions and a couple other

things we'll go into

but to keep the um papers

two per month and then to keep it nice

and not have dot one and dot two that

break across months

we inserted some quarterly round tables

and so here we are at the first one

the goals of the quarterly round table

at least for today are first to

summarize the results

of our quarterly activity secondly

to show for a broad audience what we're

working on and

doing and learning and then third

continue

and amplify our call for participation

through deep time so if you're watching

live that's awesome we look forward to

your comments and questions in the live

chat

if you're watching it in replay it's not

too late to get involved because we'll

still be around

and so if you're still interested then

we'll still be there

the sections of today we're gonna go

around

and have introductions then we're gonna

take a strategic perspective on what the

active inference lab

is and give a little narrative

information for those who might be

curious

then we're going to go into updates from

each of our three organizational units
which are
edu dot coms and dot tools and then
in future we'll just recap some ideas on
next steps and on the needs for each of
the main units of the lap
so without further ado we are on the
introduction and the warm-ups
we'll just go around and introduce
ourselves and we'll use our
regular um approach which is just first
we'll just
um say hello however we want to um give
information about herself
and then pass it to somebody who hasn't
spoken yet and then after everybody has
introduced himself
we'll um using the raise hand feature in
jitsi
will go through some of these warm-up
questions and it will be awesome to hear
from everyone
so my name is daniel friedman i'm a
postdoc
in california and uh have been
doing most of these live streams for
active inference lab
and i'm just looking forward to this
first quarterly roundtable
and i'll pass it to sasha
hello my name is sasha i'm a
neuroscience graduate student
and based out of davis california
and yeah i'm just uh looking forward to
reviewing
what's happened in the last quarter and
um
inviting people to join us

and i will pass it to alex
thanks hi everyone uh i'm alex welkin
i'm in moscow russia i'm researching
systems management school
and also co-organizer for active
inference lab
we will try to join approach of
active inference framework and systems
engineering framework
and i pass it to scott

thank you alex my name is scott david
i'm the director of the
information risk research initiative at
the university of washington
applied physics lab we're using active
inference concepts and trying to learn
active inference concepts to use in our
work on information
risk mitigation structures i'll pass it
to stephen thank you

hello i'm stephen i'm based in toronto
i'm doing a practice based phd through
canterbury christchurch university
which is sort of morphed more into a
methodological
phd as i'm exploring spatial approaches
to sense making and how that can help
theater for development disability arts
and

sort of multi-scale community
development and active inference
is actually a really powerful way to do
that
but this lab is really helping because
it's a very
new and complex area so i'm very
grateful for
the way that it's helping to really get

into the weeds as we say
and i will pass this over to evan
hello this is
i'm in russia moscow and i
i happy to be here i'm glad to that
we have the first round table and look
forward to
repeat again what we have done
last year cool
um fun stuff i i like everyone has
different context
some of us are on a little snow walk
others of us are in our cubicle
so we can kind of go through some of
these warm-up questions
either the questions being what's an
interesting or memorable experience
that occurred through active inference
lab disorder and then the same question
that we actually ask
in the discussions on the papers which
is what is something
you're wondering about or you'd like to
have resolved reduce your uncertainty on
by the end of today's discussion so
anyone can raise their hand but i'd say
one
really interesting and memorable
experience was
the entire discussion about realism
and instrumentalism which was not
on my radar it wasn't a dialectic or a
dichotomy that i had heard of or been
familiar with despite
learning about science and philosophy
and so for there to be
a sequence of papers that we work
through together including

mel andrews's paper and then following
with

um vanessa and hippolyta's paper for
that sort of arc in the literature

to be influencing our practice as a lab
by

advancing our collective understanding
about how realism and instrumentalism
worked together

and then the way that that discourse was
happening live on streams and

live not on streams and in text-based
discussion was just like very

multi-modal and interesting to see how
it kind of bends back around to actually
changing our action not just how we
think

alex

uh yeah thanks uh for me most exciting
thing

it's to meet a lot of people who joined
to our communication channels and
follow us and our activities

and starting to participate in different
kinds of work or

what lab is organizing

steven and then anyone else who wants to
say something

yeah i think one of the things that i
found really helpful i've been looking
at the perspectival consciousness paper

a bit before and um definitely had
got a lot out of it but i i had hit a

kind of a wall on my own

and then when we got into that and then
start to also

see that through our next sessions

talking about the big five

psychology traits and sort of oh okay
we're starting to get
some patterns here um so that was really
useful and particularly because i'm
there's been a big struggle with how do
i get into um
work around psychological perspectives
and then suddenly it was realizing well
this whole idea of perspectives itself
is loaded and it's kind of was sitting
there as just a de facto thing that
everything else is going to roll out
from my psychological perspectives but
suddenly
well you need to think about what it
means to have a perspective
or what it means to take a
pre-reflective more phenomenological
stance and
my gut feeling having got into active
inferences that it could cover with that
and what i'm seeing through the
the different papers in the series and
discussions that actually
is touching on these different areas and
so that's really useful
cool thanks steven and yvonne or scott
you'd be
welcome to another i think fun memory or
experience was the model streams
so um having the space to ask questions
and
having the um very generous and
attentive christopher white and ryan
smith spend
four multi-hour sessions really
walking us through this living document
that had like 25 versions and there

really were
implementing feedback and questions that
people were having and not just
oh hey you were incorrect you made a
typo but actually the feedback from the
community was like
how could this be made more accessible
and rigorous
and so we were navigating that all
together though it was the two authors
on the paper
the ones who had the most expertise so
that's sort of like a new kind of
research coalition that felt a lot more
participatory and inclusive than
other kinds of research axes that align
and then
dissolve scott
you're mute scott
and then sasha oh sorry i was muted yep
one of the things i found fascinating
i'm new very new to the
area is how um
each new thing that i learned about the
model
and the and the processes are associated
with active inference
reveals a certain new um
perspectives on existing problems and
challenges
and how some of the problems i thought
that were there dissipate
and then there's new sets of challenges
and so it's really been a
um it's it's one of the few times in my
life where i've been going into
something knowing that there was going
to be a paradigmatic shift

and i've been experiencing one it's kind
of and paradigms i always think of as
the sudden
aha this is like a slow-motion
paradigmatic shift because there's a
series of ahas
that happen so it's a fascinating
opportunity for discovering this
entirely new
framework that's out there that really
helps with the work i'm doing
and i think will help with the work that
a lot of folks are doing so that
it's kind of like discovering a whole
new world quite frankly a whole new
discipline
that is in the making and that's very
very exciting just watching that be born
it's like watching
i mean it's springtime here the
daffodils are starting to grow and i
it sounds like i'm overreaching here but
it really is like watching a blossoming
of a
of a living organism in in watching
these processes
and this community develop so i'm
fascinated by the idea that the
community is developing the ideas and
becoming
that idea at the same time and that's a
that
um intentionality and that awareness is
very different than a lot of other
organizations
that the that they're both experiencing
and exploring the experience
simultaneously

so that intentionality associated with
it i think holds great things for
not just this iteration of the of the
work and the model
but the model generally in terms of a
lot of existing problems out there
thanks cool thanks scott sasha
yeah um similar to a lot of the comments
that people have made
um yeah just been very exciting to see
the new kind of
energy and styles of participation that
people are bringing in because when you
send out the call for participation you
really
don't know what you're going to get and
you can't be too specific about what you
actually uh are asking for and so
that's been very exciting especially in
the comms
team um and thinking about
uh other ways to communicate other than
the
like weekly journal club to have uh
shorter
um perhaps kind of
little bite-sized versions of the talks
um so that's been
uh opening this conversation about
who should be doing the communication
and teaching should we wait to become an
expert
and then share what we've learned or
should be kind of like a foray out into
the unknown
and try to communicate to others what we
are trying to learn ourselves
and um i think it's the latter as we've

talked about many times in this
space and that's just a very um
exciting and scary place to be in so uh
that that's very uh exciting and
something i'm looking forward to in the
coming year
of what the participants will produce
thanks sasha um stephen and then we'll
move into
strategy yeah i suppose just
one thing that sort of bounces on what
sasha was saying there's um
the the the the foray into participation
and that whole
as a whole thing in itself which is very
vast but has been really interesting
because
it starts to open up opportunities
within some bounds at the moment but
opens up opportunities to really engage
with this knowledge
in ways that maybe you can't do in a
university so i'm actually on the board
of an organization called interchange
for peace which is community-based peace
building
which was set up by ann goodman at the
university of toronto
in the oise and she was working there um
same department where paulo freyr used
to work and
but she set up her own non-profit
charity because there are certain things
even as a
senior lecturer she couldn't do within
that role
there needed to be some way to engage
people that was he was a bit more freer

from
the constraints of the role and so she
would work with that organization i
think there's some parallels here in
like
by being able to work with active
inference more broadly
across papers across perspectives across
disciplines
may be hard to do that in a in a single
university department so that's kind of
interesting
cool thank you so here we go
into a little bit of recap little lab
101 um alex do you want to just go first
on these next
slides and just let me know if you want
you know if anything else
maybe on the next slides all right
starting on next slide sounds good so
um just as far as lab 101 here
at the end of 2020 uh we started to
apply
onft and active inference which we're
gonna get into more detail
what on ft is in a second but we started
to apply onft and active inference
to the organization of the active
inference community
as a new kind of participatory open and
accessible lab
and activeinference.org is our reference
point
you'll see a lot more information there
on
various projects and units that we're
involved in
as well as a link for the call to

collaboration and then just
that's how you get started in december
2020
we made the initial invitations for the
lab
we sent out just an open call for
participation
and set out in broad strokes the three
organizational units that we are still
working with
and then at the beginning of january
2021
was when we began official lab
activities
um mostly entailing weekly drop-in
meetings
as well as several other things that
we're going to be talking about so
that's sort of the how we got here but
there's also more to say on that
so alex on the strategy side
just pick up with our paper and let's go
from there
hey blue okay thanks
yeah as you mentioned uh we
started to apply onft
framework as we proposed in our initial
paper what was
prepared in the last year uh
where we tried uh to find a way to
find some kind of object
in reality what will help us to organize
work of our team and abroad the
community
in a way of understanding of
new ways of remote and online team
working
we believe that reality is changed

already and for now we consider
any team as online team and a
new type of
and new new types of behavior of
team members of where communication
styles should be
organized in some way to be most
efficient in this new reality
and in accordance to active inference
and systems engineering approaches we
find this object as ontologies
narratives formal documents and tools
which could be kind of informational
niche
for online team which
can structure and organize
way of doing things in the team
cool thanks for the summary first author
alex
and onft was an acronym and sort of a
thought pattern that we
introduced in this paper just as an
emergent combination of all the
collaborators working together
and so for those who might not be
familiar with each of the terms ontology
is just our structure
of knowing things and describing things
we're going to go into more detail
later today narratives are another
broad topic and our first active
inference live stream was on active
inference and narratives it's really
what drives us
is this narrative level of understanding
and communication
formal documents are referring to it
could be a cloud

document or a spreadsheet but some
document where someone says yep here's
the link
make your changes here fill out this
form that's a formal document
and then tools broadly are the software
and the hardware and the wet wear
that we use to get it done and there's
some nuance to introduce with instrument
and tool
again this is the top level onft
helps us remember that we're doing
human-centric
systems design and that entails making o
and
f and t work if you have a group and o
and f work and they don't have the tools
there's nothing to do if they have nft
uh not that nft but narratives formal
documents and tools but no ontology for
sharing their communication again it
won't work so we need all of these
dimensions to be
participatory and deep and um
this paper was written with alex as the
first author yvonne on
here sasha alexandra here uh
myself and then rj who is not on the
call but um
is also around so this was really fun
because we just put it up as a preprint
and just started going for our next step
into the organization
so we combined complexity systems
engineering and active inference
with this goal directedness where did
the systems engineering
take us towards or what did you want to

draw on systems engineering from alex
because that was definitely an area that
was new to me
thanks system engineering is a
very developed domain
of doing complex engineering systems
and great community around the world for
a few decades is working
on developing approaches what will help
to create successful systems in
different absolutely different
directions
of people's lives
so there are some set
of standards what was developed by
different communities
here is presented three of them but
actually there are
much more different papers
related to it but as
for most applicable and more most
useful and most important documents
presented here it's first is
system engineering book of knowledge
this is development of
international council of system
engineering
as for last year last version of this
document
it's about thousands of paper
of papers but it's
kind of a big document but it has
has borders of the domain and
if people interested just going through
outline of this document
we can understand what is
it about and how it could be
used in different directions so there

are some set of
iso standards here is present in
15 to 8 it's for
mostly for systems life like
life cycle processes and defines
processes what happens in the system and
in
all type of system what also need to be
created and developed to
make to produce some system of interests
and uh
object management group has own
standards as
essence is initially was developed for
software engineering but now
we have understanding that concept and
schemas from
the standard are also useful to
to apply to the works of the
remote teams cool
so a lot of resources that are
professional
and also modern and global
from systems engineering one piece that
really
bears upon our system's design is this
concept of a system of interest
related to attentional regimes from the
ontology of active inference but the
system of interest is the engineering
terminology
and our system of interest is the active
lab participant so that's what we're
trying to
instigate change in here we call it a
nes
change non-equilibrium steady state
which is another active inference term

and that's just referring to this idea
that the steady state operation
even if it's far from equilibrium of the
participant
is going to undergo a systems change and
an active inference there's two ways
that a system can change to reduce its
uncertainty to minimize its
expected free energy the two ways it can
change are by modifying its internal
states
which is learning and development or by
modifying its external states
through action and niche modification so
we're thinking about
the system of interest as the
participant that's a person
who's starting out indifferent then
something
changes that next change results in the
next stage of this person's
progression along this axis towards
being an interested person
somebody who's following the active
inference lab towards somebody who's
participating in the active inference
lab
and where systems engineering and the
idea of enabling architecture or
enabling infrastructure
enabling systems comes into play is for
this
ness change from somebody who is
indifferent or unaware
to someone who's interested there has to
be an enabling system
that enabling system in this case
there's other roads to take

is an operational live stream if it's
just a conceived live stream but it's
not at a stage
in its progression where it's actually
producing then this person is not going
to hear about it so it can't be enabling
infrastructure until it's operational
and then
how does an operational live stream
occur well that's
kind of the fun and that's what we've
been experimenting with but it turns out
that the dot-coms unit
is the group of people who with enabling
architectures of their own
are the ones who enable as well as
fine-tune an operational live stream
which is again what enables a certain
type of change to occur
in the system of interest that's the one
that we're focused on
that's the one that we care about
changing and everything else is a means
to that end
alex what else would you add about that
from the systems engineering perspective
um i just want to mention this this is a
small part of a big picture of this
enabling chains which should be
developed in system engineering way
and here it's
presented explicitly that for any
uh any enabling system there should be
uh some kind of in our enabling system
and from
on the bottom there is this one blue
blue arrow it's uh from another
enabling system in the chain which could

be for example
obs or youtube channel
exact tools what i enabling work of
dot com's unit which is producing
operational
live streams but also dot coms unit
is producing other types of enabling
systems
which can be used on different
non-equilibrium steady state changes
points
so it's kind of very complex
structure of enabling systems but
without
managing attention of
to to create it we can't achieve
final needed results on changes in
somebody's mind
as we propose here and i should to
mention that
initially this approach was developed
in systems management school in moscow
so we adapted
for our lab but it's also based again
on that's international standards what
which are developed by much more
broader communities of engineers so we
are trying to
use and apply state-of-the-art
enemies domain for what we are doing
any thoughts or questions on this yep
steven go for it
yeah i just thought i'd ask um you're
using the non-equilibrium steady-state
change there so you're introducing
something
into that and uh how how's that um just
maybe you could speak to that i mean

because it gives you some other options
potentially than saying just behavior
change
which may be used in i don't know if
that's what you use normally
so maybe just speak to that and how that
might
pose challenges for you as you're trying
to work it out or
opportunities and just where you are
with that kind of thinking
thanks for the question so there's going
to be many
answers and in the end it's enacted it's
how it actually plays out but
opening the door to calling it
non-equilibrium steady state change
let's just say you want someone to brush
their teeth every day but they're
currently the kind of person who's not
brushing their teeth every day
yes it is a behavior change it's an
action change but
it's going to be a complex non-linear
dynamical system the person in their
environment
all of the richness of that is what
needs to be
worked within so if it was a dial and
you could say make it louder and just
turn the brushing per day from zero to
one
well yeah then that is kind of like an
equilibrium steady state it's a
knob that's just resting in a position
and you just make some sort of
perturbation and then it just stays
where it's going to stay

but for the systems that we care about
which are
also the systems that we hope are within
the scope
and the potential of active inference
modeling which are
embedded inactive complex agents
we want to make it clear that we're not
just talking about
environment change we're not talking
about behavior change we're not talking
about their mindset changing
it's about the total scenario
developing in a non-equilibrium steady
state fashion just like an ecosystem
would
towards something and so then that helps
us be really clear that we're bringing
that richness to the table
but we're going to be talking about it
in a way that's human-centric
and progressive in terms of incremental
changes
to our system of interest and then we
can cope with how many um bifurcating
dependencies there are and there are
many including some that aren't within
our control
and it just helps us keep it organized
and kill two birds with one stone by
communicating better by using active
inference ontology
yeah quick follow-up and then to scott
yeah just saying yeah i like that i
think that also
gives a nice segue between um
the system of interest that you're
working with

and the potential unknowable
impacts without saying without getting
the messiah complex
but the idea that there's there's the
the work
in development work where there's
something called outcome mapping and
systems change which is you can never
know because it's
beyond any ability to know it might be
interesting that use of non-equilibrium
steady state
as it it seems to give a bridging point
because at the moment they just use
behavior change as well
so i was just saying to know that could
be interesting thanks well we certainly
don't know the effects of our actions
because the butterfly effect so just
breathing you don't know so
right outcome mapping is a summary
statistic or it's a measurements
within a non-equilibrium steady state or
a change to a system of interest
scott you know it
does yield an interesting change in
terms of
consciousness and cognition
because you know one of the problems in
consciousness is the mind can't know
itself that whole challenge
and so you have this kind of a feeling i
don't know if it's
been stated this way but kind of a
guideline incompleteness
that you're going to ask questions
within that system that won't be able to
be answered from within

the system in this case one of the things that's really fascinating to me is it is a it seems to be a comprehensively introspective mind but not to mean it's not expected extrospective also but uh outward looking but it's um the the i guess what i'm leading to is the thing that's interesting about active inference to me is that the system can know itself in a different way than a system that doesn't know that it's performing active inference and so a self-awareness of active inference both in human cognition where you've had this perennial challenge of can the tool know itself and here you have the tool and we're using a tool to deploy to understand knowing in different systems and it's kind of interesting because we can construct up systems that can know themselves to the extent of the active inference model and so it feels to me like it's going to reveal a lot of other unique aspects of thought and consciousness and again maybe it's just a model like different models of you know the mind is a robot the mind is a computer the mind is you know whatever is historically relevant then maybe it's just another model but it feels like this one's going to reveal some process elements

that will have um provide us as humans
with the opportunity
to understand connections to other
things that
also rely upon and display those same
process elements
so people to if and to the extent people
can start to think of their minds
as more like a beehive their minds as
more like a
tree and its relationship to the fungus
and and
forests that feels like it will be
of service to our resilience and
sustainability
those new perspectives thanks cool
so let's do last little two slides on
strategy
thinking about active inference lab in
terms of
internal structure and also its niche
its external interfaces
and the idea of interfaces is central to
active inference
with everything about the markov
blankets different kinds of blankets and
interfaces
as well as computer science and
engineering so it's kind of
something that is almost pre-adapted to
work as far as our current
lab structure as stated we have three
organizational units that we're about to
spend
time going into updates4.edu tools and
comms
then at the interface of internal and
external are the deliverables

that we're looking to make real change
and progress on
so the body of knowledge is a
deliverable in a project that's
maintained by
edu that we're going to get to next and
it's
of course a work in progress they all
are dot-coms
is the unit that brings out the live
stream
and people can be on multiple units or
be different roles across different
areas
and there's connections between the
areas and um
comms is also the unit that interfaces
out with
broader communities so looking at the
bottom here
there's sort of a tri-color scheme so
blue
are the broader communities the broader
communities of
science learning and doing are everyone
so we can also
call it just whoever is within the
sphere of influence of the world
then there are those who are learners
and practitioners of active inference at
any
level of familiarity or of any
background
or career trajectory just like we have
complexity learners complexity curious
and
people who are just curious and know
what the word is you're already on the

path

then there's the lab itself which is

just one

organization in a bigger situation with
other people who are studying all kinds
of

amazing stuff active inference and not
and

comms we imagine will have sort of
several different ways to interface with
different communities

one is just by connecting with the
active inference learners in that
community

so here is um the basket weaving
community

and there's a a special interest group
of basket weavers who are interested in
learning active inference and that's how
we interface

and then um one could also imagine that
there are communities

perhaps that are more science oriented
but also as we're seeing potentially
performance oriented service oriented
social change

finance all these areas where there is
a conversation happening with dot coms
as our interface as well as a broader
community uptake and effort and interest
in learning

so this is sort of our internal um this
is sort of

internal structure in light of the niche
and then just to sort of zoom in

on the internal structure then anyone
can give their last thoughts on this
strategy before we go into the project

areas um

the initial stage is attention without
attention to active inference and the
lab

there's nothing to latch onto nor follow
after attention we think about
accessibility

and about onboarding because if it's not
accessible

and there isn't an onboarding there's no
one on the ship if you can't onboard
onto the ship there's no one on it so it
has to be accessible and in terms of
onboarding for us

that's o-n-f-t that's starting to learn
and use the ontology of active inference
learning and understanding and
co-creating the narrative of the active
inference lab

the formal documents like the forms and
the documents that we use to organize
everything the calendars

if you're not on those then you're not
on board

yet because that's actually how we work
and then there's the tools which is
similar with the formal documents but
the tools

are the software and the hardware as
well as the wet wear and the behavior
and just the practices

that allow one to be a participant and
then there's this continuum of
participation

where we just take a role-based approach
so if someone is a beginner
in learning something we'll try to find
a space

where they're going to be able to learn
by doing and assisting
and that's what participation will look
like for that person at that time
so um any thoughts on these sort of
outlines
yeah steven go ahead yeah
i think that um once it's nice to think
about the onboarding with the onft
to sort of gives you a way to break down
um ways to help uh
understand what bits you people are
trying to be under
um trying to relate to and i think then
once there's that participation
and wherever that can go you start to
get into the realms of sense making
and different forms of sense making that
might be happening
once you have started to gain that kind
of
um framework of understanding and
interaction and seeing where people are
taking in different ways
so i think that's quite interesting
um scott yeah the other thing i think is
interesting here is
in making the um the journey explicit
and mapping it for people everyone is
somewhere on the map
everyone who doesn't know anything about
this is a pre onboarding person
you know some there's some location for
folks
we think it's really important the other
part is one of the things that i've been
wondering it feels like what's going on
now in the world

is that people are moving from data
integrity to meaning
integrity questions so i could say data
security to meaning security
but i mean it more broadly and so this
process really falls well
into that because the data is much more
looking at the
nodes from a network theory graph theory
kind of perspective
and this the meaning is looking at the
relationships and the edges
and because this whole enterprise is
relationship
based in even in terms of the current
state of a node
um i feel like it offers an introduction
for folks to look at the
really the more dynamic elements of it
and so complexity
you know you don't have to talk about
complexity for this to be useful right
people can go into this and
develop awarenesses and realizations
that help them cultivate
meaning integrity and meaning
understanding
uh in the world without knowing that
they're doing complexity
and i think that that the the fact that
this can be helpful to people
um to understand the dynamics of their
relationships and the dynamics of
internal dynamics of other entities with
which they're relating
feels to me like it's going to be very
valuable in this next stage
of security and privacy liability

um limitation all these things where
people want greater integrity of their
interactions

it feels like this practice and this
awareness can help result in that
thanks thanks um okay then we're gonna
be talking about these projects and our
progress

so stephen quick no
okay okay sorry okay so nice up sasha go
for it

i just wanted to say um one thing so
there's

there's three teams and i think at the
start

maybe there was a concern that the teams
would be kind of siloed and
maybe working on their own projects and
not

linked and so i think it's been
surprising

um in a great way that the teams are
actually very interconnected because of
the people who are on them
but also because the teams are dependent
on

each other for content and for
development

and so that's been a really um i guess
exciting and pleasant surprise that
um the teams are going to continue to
work together and kind of bounce off of
each other

in a way that um really isn't siloed or
sequestered

um as i maybe wrongly expected
cool and welcome dave hello

yep and tools doesn't have any links out

at this point but that's just a current snapshot so we'll update things um here we go with the update on the edu organizational unit so the edu the goal of this unit is pretty broad it's to create a participatory and dynamic active inference body of knowledge and our primary areas of progress have been related to ontology development specifically a terms list which we're going to get to in just a second and the next steps for this section which again we're going to go into more detail when alex describes the ontology we're still looking for feedback on the terms list and we're making progress on ontology development and the um ontology life cycle but we're still looking for ontology experts so people who have worked in an advanced capacity with ontology so alex maybe you want to take a first pass on the active inference body of knowledge what is this diagram on the right and then what does it mean for active inference okay thanks yeah on the right is a diagram from it's about ecosystem i guess which is grown around international council of system engineering and well there are a lot of areas of people's

activity
and all of it we can see
is based on
system engineering body of knowledge so
when you have
such kind of body of knowledge
you have option to create
different educational courses maybe
certification programs
maybe a lot of different stuff
to people to work with to
learn to develop themselves in the
domain to communicate
when you have some kind of knowledge
artifact which you can refer
and which have its own life cycle
but it's kind of
common agreement of uh domain
understanding
between the community and this is a
cornerstone to develop
different type of projects and
activities
and our hypothesis was could we use the
same approach
to develop active inference body of
knowledge
which should also provide us much more
importances
to develop the domain
organize the work of people and make
progress in all possible meanings
cool yep it was this idea of a body of
knowledge which is implicit
in every developed and mature field but
the engineers take it to the next level
and the systems thinkers working
together unsurprisingly

so it's like very cool to see how
there's really a specific relationships
between different parts here
these are actually controlled vocabulary
terms like in forms and drives these
aren't just like sort of synergy type
words and so by defining the interfaces
and the progressions it actually helps
it
be participatory and accessible and so
we just were
motivated by active inference one day
realizing a truly mature body of
knowledge
which isn't just a textbook it's not
just an unpacking of a paper it's not
just a journal club discussion live
stream
it's actually formal courses
certification
standards but also practices that unify
our community so that will be really
awesome and then it says what else
because
we just listed a few english words but
this is something that we're going to be
enacting together so that's why
one option is you know the live chat
that's in affordance for feedback
for those who are on the stream yes it's
an affordable feedback and for co-design
and then it's the actual doing and that
will decide how it is
so this is going to be pretty cool to be
working on with
edu and then we have a lot other things
to say in ontology but stephen go for it
just like to say i think it's quite

interesting the um that bit in the
middle the boundary of systems
engineering so i quite like
the uh you know with the participatory
nature and the sort of
evolving nature of uh sort of
discussions you've got
it's like the right um the right hand
side there is is is the
the more kind of structured formalized
way it's bridged by this boundary of
system engineering with the
system engineering community so it's
kind of
be interesting to see how that nature of
the boundary of system engineering
um relates back to the nests
non-equilibrium steady state and whether
there's some interesting dynamics there
which may be a similar or different to
what's done in other areas of uh systems
engineering
cool so um as far as progress on our
journey
one of the key insights that many people
don't expect
but then once they start understanding
the sense of the word more broadly
they start to see is that ontology is
the root and the backbone
whatever metaphor you want to use it's
the structure it's the clothes hanger
it's a formal and explicit specification
of a shared
conceptualization of a domain that
doesn't mean it's fixed in fact it's
dynamic
and participatory and then ontology is

communicated between
people as well as computers and that's
what allows us to have heterogeneous
and distributed systems and this is a
paper um i think by its ontology
learning from text
by i think yang at all or i'm not sure
and um we could look but
go right now and it's a continuum of
formality of ontologies
so not every one of these terms many
people would associate
all these things with ontology but we're
taking a broad perspective on ontology
and we're starting with the simplest
which is a terms list which we're going
to show in a second
terms lists can be updated as well as
new terms can be added terms can be
removed but
the richness of the ontology which is uh
like a sort of way that things are
related to each other
the richness can be developed to
initially include definitions and
synonyms like a glossary or thesaurus
then looking more like a database like a
data approach
then all the way out to formal logics
that allow
almost computation in an analytical way
on
knowledge relationships which is
extremely powerful but usually it's
quite domain specific
so we can't put the
cart before the horse i was thinking put
the inference before the active but

that's fine

both directions are fine so we're

developing dynamic ontology life cycle

and the stage that we are at in this

first quarterly update

is that we did complete the first stable

version not the final version of the

terms list

that's the active inference lab terms

list v 1.0 candidate

and we've been getting several weeks of

feedback on the terms list

improved it a lot also with very much

appreciated feedback from

carl fristin and this terms list is not

always going to be a spreadsheet it's

not always going to look this way

it will develop into richer

computational resources

that allow us to do translation allow us

to make

educational content at different levels

to connect formally to different fields

to think about how terms are being used

across different spaces to provide

points of contact

because one thing that we noticed when

we were making the terms list in the

supplement which is sort of

words that we put in the b tier was the

b tier words

were like a lot of isms and theories and

ideas and

ways of doing a certain mathematical

formalism and then the terms

for the actual core set that we will

be focusing our language use and

understanding on

they include really um powerful words
like
accuracy action agency ambiguity
attention that's just the a
section and there's less than 90 terms
right now
so this is pretty fun and that's
basically where we're at with
the ontology project which has been the
main focus of the edu
unit and the next steps are to continue
up this progression to move towards
better understandings of our ontology as
well as to use
text analysis type approaches to
understand how these terms are being
used
and provide a little bottom-up or a
data-driven ontological approach
anyone have any thoughts or comments on
ontology or edu before we go to the next
section
all right let's stephen go ahead
yeah i'd just like to say this is really
very very useful
um actually i think even
the process of people um you know going
through these terms they could be an
interesting educational piece
um just like having people explore
how these terms arise for themselves you
know and actually
maybe even gamifying some of this
because in exploring and finding out the
terms you actually
understand an awful lot more about
active infants so it's it's
it's good on a number of levels actually

cool
and also dave thanks a lot for your
really
powerful insights into ontology and just
language and helping us think about that
in a in a
structured way so the second project is
comms second organizational unit because
there's teams there's projects inside of
a unit so for now it's sort of like the
unit and the team are really similar
but when the organizational unit is
larger then there will be
disjoint teams and projects inside of
comms
so the goal of comms is sort of two-fold
the first goal is to carry out all forms
of communication with external entities
and the other goal is to organize the
lab's internal projects and activities
so it's internal communication and
external communication
because when we're thinking about
systems design for online teams for
remote teams which are all of them now
those are what we have access to
internal communications design
and then external communications
interfacing
and towards that end i think comms has
been
a fun experience for a lot of people
we had several projects within comms
so again with overlapping personnel at
this point but
um not for always there um
several uh series that we carried out
so just to sort of describe and

distinguish them the active inference

live stream

is the main weekly discussion that we've
been having

it's publication centric other than the
quarterly roundtables

it's like we discuss a paper and that's
why we have the calendar that references
the paper and we schedule things
a lot in advance for this one and for
each

number of paper like number 16 is going
to be the same paper for 16.012345

however many sessions we need on that
paper and the idea is that dot zero

is like a contextualizing a
contextualizing video

um blue as a amazing contextualizer

and we'll be doing 17.0 tomorrow um

so probably pull an all-nighter or
something on that one

and then the dot one and the dot two

videos are participatory group

discussions ideally with the authors
around

also be cool if authors wanted to help

us with dot zero but the idea is we can

invite and really honor and respect

the authors by having them show up to a
discussion

where there's going to be a group of
people people sharing their perspectives
at different

stages in understanding and then it's

all good for participants to have

not read the paper or read the paper a

little bit just circled questions

highly added a ton of notes it's like

wherever people are at
the authors are the ones who put out the
paper and so that's been great to talk
with the researchers in the active
inference
community then um that sort of led
to a desire to go a little bit more
technical at times
and ryan smith and christopher white
kicked off our model stream series with
four
model streams and the model stream is
intended to be like a tutorial or a code
walk through
or more like a deep dive into modeling
because then we can be really clear
where we're doing a how-to on modeling
and then
in a live stream uh regular format would
be more about
like when we had alex chance for octave
eight um
that was talking about implications and
asking some more qualitative questions
about the project
and then the new um newest series is the
guest stream
and majeed benny came on for the first
guest stream on feb
24th and the guest stream is just our
grab bag
our wild card it's like if a new paper
comes out
we could have a discussion with the
authors 48 hours later
if they're available so a guest stream
the idea is on
short time scales or if someone says

yeah i'm really busy i could come on and visit in seven months then we can also account for that so any time of day or any availability that that somebody has who wants to be leading a session or be a guest on a session we'll just put it in the guest stream component and then there's bite-sized so maybe blue would you want to say anything about bite size or any other projects that you're thinking about on this front sure the um bite size really uh you know i have a lot of people like where i'm like i'm doing this live stream it's really cool you should come like participate and here's this big paper like the paper that we're doing for the 17.0 is like you know 30 pages long like here read this paper and join us on our live stream and people get a little uh intimidated by that and so they don't really want to engage at that level but i've definitely had a lot of feedback from people saying like oh you're gonna make it into like an audio podcast like short like 15 minute little clips that we can listen to on the way to work or while i'm out for a walk or whatever and so i've had a lot of interest in that and um i think i'm i've got two episodes already clipped from like i'm starting at the earlier live streams and moving up um

and so so i've got a couple episodes
that already clipped and ready and one
more that's almost ready
so i think we should be launching like
as soon as i get them
pushed into the rss feed maker we should
be launching maybe as early as
next week probably so just you know
short little episodes for people to
um listen to and um you know that that
are maybe not
don't involve reading a 30 page paper
and doing a bunch of math
uh but still people can engage at a
level that's kind of
um thought friendly and right commute
friendly
right awesome yeah thanks a lot for
taking the initiative and helping that
project happen and
we'll probably make another youtube
channel like
acting clips or something and then
we can figure out how to make it work
and we'll keep everyone posted through
our
communications channels so also another
deliverable with the next steps for
comms we're going to continuing continue
working on our
series that exist so the live stream has
a calendar
um that's in the info box for each video
the model stream
we're in conversation with several
people as well as for guest stream with
future dates
so that's how it is with live stream you

can schedule it but then in the end it's better to tell people when it goes live or after because you never know what's going to happen last moment so we just um are looking to schedule and coordinate with new participants of any background or interest if somebody wants to give a model stream or you know be the one who's walking through their tutorial or their blog that's awesome and we have so many people who have written like their own sort of um free energy or active inference manifestos or learning documents so how to surface that or who could we invite in an adjacent community to do a guest stream or like an interview or could we do a special other stream could we do a math stream or a kid stream or just joke stream who knows and then we're looking towards the kinds of communications that are enabled by the markov blanket of a remote team we can send data and that data can be text like a blog or a wiki or article it could be audio only it could be video only it could be a still image it could be a physical object so what are our affordances as an organization and then there's some that we can't

afford

to do now but maybe we'll afford them
later

and then just the last note on comms is
it's been pretty

interesting to work on what basically
amounts to a live stream organization
and implementation checklist

it's just a sequence of commentable
google slides

for now but it partitions

some of the roles of a live stream just
one perspective

in a way that makes sense inc it

includes a couple of things

that people might not expect about a
live stream for example that what they
see on their computer is what other
people see or that noises

might be not for them but other people
can hear it or vice versa

and um that helps us make sure that we
have a high quality live stream
experience for everybody

as well as an experience that is
accessible because someone can

um reduce their anxiety reduce their
uncertainty about

the who why where when and then there's
a total

um direction to continue in

with just how to communicate and
dialogue

so i know it's something that many
people on this

conversation and and in the lab think
about a lot like

whether it's in the context of

transformative dialogue
productive innovative whatever adjective
is meaningful to you there's something
about improvisation
and coming together with the affordances
that we have
to just hopefully do good and have fun
so how do we make that possible
has been what comms is about clarifying
and communicating on
any last thoughts on comms before we go
to tools
um yeah alex then stephen
yeah thanks i just want to add uh
in relation to checklist that it is
a checklist is one of the main concepts
of
and approach from systems engineering so
we are trying to apply system
engineering
approaches for actually
all types of activities what we are
doing
cool stephen yeah i suppose the
the idea of the niche or creating as
an awareness of the niche and the
affordances in the niche that we're
giving
people as well as their their own sort
of personal
journeys kind of a nice way to think
about this so as as
as the list is being ticked off it's
like
connect with self and participants to
bring the relevant energy
so that's kind of a bigger term
than you might often get in an

engineering context but you know that
that's
but that is the sort of thing that often
gets forgotten you know especially if
you're
so it's good to to like have that as a
yeah let's just check in
you know and just get that chance to
make sure that
we're in the right headspace because
this it does take some time to
to reorientate so that's quite good yep
and
the the challenge and the opportunity
are
two sides of the same coin for the
remote teams we can have international
conversations and so yes maybe you know
dave's video blips out
once in a while or somebody gets kicked
off but it's like we're having
conversations that are very special
so instead of relativizing it to some
imagined reality where like tech is
perfect and free and everyone has it
we can reference to what's possible and
then we work with what we have and make
the best of it and
some people their rate limiting step to
live stream participation
is psychological or internal whatever
for some people it's um logistical they
can't
make the time work maybe they live in a
different time zone for other people
it's technical
they have the time and they want to do
it but they don't have a camera or a

microphone or
internet connectivity or something like
that so when we
have a role-based approach to
participation not like
live streamers not a title in our
community it's like
who's a participant in this live stream
today
in quarterly roundtable one which role
is everybody playing
and we actually have an answer for which
people are playing each role
and whether they did or didn't do the
checklist exactly today
probably few did um i know that the
broadcaster didn't
but that's all good because it at least
is there
and then we can iterate on it and that
will i think really
be powerful and effective okay any other
thoughts on comps but yeah we're just
looking forward to
learning by doing and communicating all
right tools so this is kind of a fun
section i think the goals of the dot
tools
unit are again sort of twofold an
internal and an external
view so the internal goal of
tools is to enable effective tool and
instrument use for all active inference
lab processes
so we need to figure out which tools
will work best for
onft for onboarding and participation
otherwise we don't have

anything but especially as we
build more scaffolding for
our lab and stabilize some of the tool
use inside of our lab and have more
regimes of attention to spare
on external views we're interested in
exploring and designing
affordances within our niche which is
all we can ever have
resulting in effective action externally
facing
and also innovating in terms of tool
development so
um tim and others who we had such
great conversations with in the weekly
discussions that
relate to our progress um and several
others scott and schneider
and many just awesome people who came
through there's like
us using tools to make the lab better at
what it does
and then there is the tantalizing
and exciting question of what active
inference driven
tools look like whether there's an
active inference computational
or software element and so the tools
look like
slack or look like email or look like
something that people are very familiar
with
but it's like a recommendation engine
that changes
how things are done in some subtle way
so whether we talk about
implementing active inference on top of
the tools that we know and see today

or whether we're talking about truly
novel
and unexpected ways of active inference
and systems engineering and systems
design
coming into play in the true um
imminent future of online work
that is where we want to go but um
with a couple people and a lot to figure
out internally
we haven't gotten there but like people
have been saying
all of the goals and units are
intertwined
and to use a figure from our paper the
fiatkin at all paper
we're taking a function driven approach
to thinking about this internal and
external team communication
and the tooling for that using the
active inference framework
because designing the system for the
physical
office you know you need the hammer to
build the hallway so that you could have
a certain
connectivity of conversation or certain
architectural suggestion
but the hammer our tool is actually the
modification
of communication pathways communication
being included
data being shuttled back and forth
between parts of a computer
or across the internet also a part of
communication between non-human agents
so we sort of even
from last year wanted to use the idea of

a markov blanket
and blankets or however they end up
being defined with interfaces
as sense incoming and actions outgoing
and so we have
internal team communications and then
external team communications
which are all important sense in action
internally and externally just like
multi-scale nested active inference and
then tool development is about making
that happen
so we're going to run now through the
tools
including ones that i hope some people
will be familiar with maybe some
people will not be familiar with and
anyone can just raise their hand
if they want to mention something about
tool so
the single source of truth from a file
sharing perspective
for the lab is a google drive folder
from activeinference@gmail.com
and this is our activelab folder we have
the three
organizationalunits.edu tools and comms
and then we have dot admin which is the
uh
minimal administration for the lab and
just like the form
responses that are not part of one of
these other
organizationalunits.edu but by and large
everything related to these projects
is totally viewable and if people RSVP
to participate
they're added as an editor to the folder

if they
um just want to see the link then they
can comment on everything so everything
from the slides for all of the
presentations to the feedback
for the podcast are in each of these
folders
and uh it's a participatory and
hopefully an open way
to organize so any project that people
are hearing about that sounds cool
or something that they think is an
adjacency or a role that they'd like to
play
even if um just a very
vague sense of what it could be just get
in touch with us because basically
this is our current project
to go into a few specific areas
in our work on the ontology and terms we
used two different tools
we used google documents
as we used with many projects documents
and sheets but then also we use protege
which is a online the web protege is an
online ontology editor
lightweight but pretty powerful and
there's also a desktop application
and this is for ontology construction
and sharing so that's pretty fun just
learning about ontologies and different
ways of viewing
um terms we
use um gather.town for spatial video
chat
and um right now we're using jitsi which
i actually didn't take a screenshot of
we could

do it total uh during add but
gatherer.town
is allowing for interfaces uh
and affordances like movement while i'm
taking a screenshot stephen
just maybe give one thought on gathering
active inference and where that might
come into play yeah well
it could be interesting i think active
inference and the nature of how
we relate into our affordances it's kind
of interesting when you start to be
sitting at a table as an avatar
and how that feels slightly different um
to how you are maybe when you're sort of
in an open space
discussing aspects of active inference
so it might be it might be quite
interesting as we
start to explore how some of the active
inference processes
are ontologically and then how they are
when we start to experience um
stepping into the dynamics of
active impress inference because it is a
a process
theory rather than a static
sort of diagram telling us what the
world is so
that opens up some possibilities um and
uh yeah i quite enjoy
meeting everyone that way so i i don't
know how others find it but
i think that's uh yep there's
possibilities
yep fun and jitsy is a little bit more
standard but it's an
open source video platform and we have

an aws server
so it will cost a couple of cents or a
little bit more per
meeting but then we know that we have
the reliability
but also meet.jit.c is free okay
then for text and file sharing and the
official
organization of each of the lab the dot
admin
and the three units that we have as well
as dot public on key base
this is um a platform just it's not
about
the pros and cons of any of these
specific platforms we're just saying
what it is that we're
using but key base is a really great
platform at the current
moment for doing the kinds of things
that we want to do
in the back end for official
organization
this is like the internal communications
of the lab then there's the external
communications in the lab
now it being an open in a participatory
lab
sometimes it's a little bit hard to say
where is their internal and external
communication because
all of the documents are visible all of
our teams are joinable
so where is the line between internal
and external communication
basically there's communications that we
address to participants and
members of our lab people who are

actively um somebody who would just directly solicit feedback to in a sort of off-the-cuff way like somebody you might bump into in a co-working situation ask for feedback whereas external communications we try to communicate um sporadically but regularly and uh well structuring our external communications like having professionalism with our newsletter and with our live streams whereas just popping on a video call with the same people it'd be an internal lab discussion because it wasn't being performed externally but they're ultimately all open i guess so for external communication specifically but also increasingly project collaborations just ones that aren't the three organizational units of the lab which is our focus as organizers is the actual organization of those lab units and incrementing those systems and including participation in a lab and then people are just so amazing and creative and have so many awesome project ideas and collaborations so in the discord i know that we'll have papers and research and projects together that are maybe even only constituting our lab participants but will not be part of the exact

organizational unit
just be people collaborating we have a
discord
and the join link is probably in the
video description and
live chat and here and then there's
twitter inference active
on twitter then um for live streaming
uh if you're watching it it's on youtube
and then if you're
live streaming it then there's this
dashboard for youtube live streaming
and then currently as broadcaster i'm
using the open broadcasting studio
obs project.com and then just that's
what i'm
using right now but the fun part and
just to sort of call back to the roles
of comms is the um the
facilitator and the event organizer in
the broadcaster the participant
it could be one person all doing these
roles for one person
or it could be many people with these
roles and it could be split up across
different expertises and skills
and then lastly aweber is where we have
our
monthly newsletter so there's been two
of them end of january
end of february so far because now it's
march 2nd
and that's going to be just for updates
on our projects and sort of sharing
similar information
to this live stream but just in a
bullet point project by project way okay
any tools thoughts that's sort of the

active lab

stack or just where it is but we know
that some of these platforms are going
to be
closing or changing or new ones come on
to the scene and

so yeah we just are kind of at the spot
we're at

and then as the affordance becomes
salient and preferable while also being
aware of switching costs then

we'll do what we have to do to scale
any other thoughts on tools though oh
stephen go for it

well i was just wondering if there's are
you actively looking for any tools to
bridge any gaps at the moment or you
sort of you feel you've got enough at
the moment like is this

i'm just curious if that's something
that you're trying to look at

i would respond in terms of optimal
experimentation

and optimal foraging on a rapidly
shifting

affordance landscape there's tools that
are just at

working quality today or in six months
that a year ago

we hadn't heard of so by staying
connected

to people with various domain specific
expertise and kind of

use specific expertise we want to stay
attuned to trends

on the tool landscape and then dot tools
is the spot where the people who are
the most interested in that will come

together

to look for internal opportunities for

local and global improvement

like fine-tuning the discord channels or

going to a different

platform for that function so it's a

need-based

engineering approach and then also

once there's again a stable working

version or it seems like the global

reorganization of the tools internally

is off the table for some period of time

or something

then there's more attentional capacity

for external collaboration

okay hashtag future all right

so to kind of recap the next steps for

the lab

the main next steps for the lab are the

next steps within each unit

which we can talk about with i guess we

can just say the edu is

working on improving the terms list and

on

developing the terms list down the road

of ontology

formalism dot coms is interested

in continuing and expanding the current

series which are the

weekly live stream the model stream and

the guest stream

and then also opening up new series or

types of content like audio only bite

sized or

who knows other media that are possible

with tech

and then in dot tools with a two-fold

goal of internal and external tool use

and development
they're currently assessing
needs and um
seeing what can be done as evidenced by
these discussions we're just having
today
as we're heading this is from the point
of view of the admin
as we're heading into our second quarter
of activity we're
looking to focus more on preparing
detailed narratives of active inference
lab
and for participants to help
structure the end of onft
because we're starting on ontology with
the terms list formal documents are
related to tools and we have what we
have and again
we'll locally and globally fine tune as
we need to
and then the tools is for now
the formal documents and communication
channels but that was the whole point
of our paper in 2020 which is for remote
teams and communities
communities of practice however you want
to think about it the affordances for
design
are the ontologies and the formal
documents
because those are the tools that the
team uses
for communication and then what's being
communicated
is heterogeneous it could be text it
could be a
image meme it could be an emoji it could

be a link to something
it can be very very specific pieces of
data
that are getting interpreted in the
context of what the narrative which is
the end part
so o f and t are kind of like the
observables and the systems that we can
actually increment and focus our
attention on and
narrative is something that you can get
meta about and focus
on but really it has to be enacted
and changed through non-equilibrium
steady state
type design not equilibrium steady state
type design
so the reason for this narrative
preparation which also alex
really led the way on with laying out um
participant life cycle
from experience with the system school
we want to
structure the access and
interpretability of lab materials
so people can have their participation
catalyzed so the rate limiting step is
it that they didn't know where this
spreadsheet was or they
solid spreadsheet but then didn't
understand how to get in touch with
something to schedule this time and then
um
the needs for the lab are in the
edu project any participants who are
interested
in ontology and uh education making or
taking coursework

formalisms learning by doing teaching

dot coms

is always looking for participants

looking in uh to areas like writing

teaching connecting audio visual

production

and events co-organizing then

tools we are looking for participants

who are interested in technical

development

computer science user experiences

toolkit

use practices

also accessibility and how tools are

used across different communities

and then the admin are participants who

are interested in

topics like logistics education

communication

tools these are the projects hint

community mentoring and service so we're

just

looking to um our diagram

and to how we can increment people as

our system of interest along this

continuum here

in the context of this evolving

structure

and that will be cool any thoughts on

next steps or

needs for the lab

the last questions are just um kind of

the questions from the regular papers

just

on the inference side what are we

curious about learning more about which

is

kind of an action oriented curiosity

and then how are we wanting to apply
active inference and then all the
questions that we ask about
papers but pretty fun thanks everyone
for the first
quarterly roundtable
pretty fun any last thoughts otherwise
that's good
alright so thanks for participating and
see you later bye